

Deliverable 1: Complete Requirements Table

Course: Software Engineering Lab 1 (PES University, Dept. of CSE)
Student: Shashank D | **SRN:** PES1UG24CS433
Problem Statement #05: Academic Elective Bidding & Allocation System

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately (sum == 100) and prerequisites validated. Fail: Total allocated credits != 100 or unmet prerequisites bypassed.	Ensures fair bid distribution aligned with student priority and academic eligibility.
FR-002	Functional	The system shall allow the Academic Registrar to configure elective course offerings, section capacities (min/max seats), and designated timetable slot constraints.	High	Pass: Course records saved with designated capacities and slot times. Fail: Duplicate slot assignments without multi-section flag or invalid seat caps.	Establishes boundary conditions and constraints required by automated allocation solver.
FR-003	Functional	The system shall execute the multi-criteria optimization solver to allocate courses based on bid weightings, CGPA tie-breakers, and room/slot collision avoidance rules.	High	Pass: All eligible bids evaluated; seats filled up to max limit with 0 clashes. Fail: Student assigned to overlapping lecture slots or oversubscribed course.	Core algorithmic automation eliminating manual clash resolution.
FR-004	Functional	The system shall allow students to view real-time provisional seat demand percentages and adjust/reallocate their submitted bidding points prior to the deadline window closure.	Medium	Pass: Bid updates reflected instantly in draft pool before lock time. Fail: Modifications permitted after deadline timestamp.	Provides transparency on course popularity, allowing students to strategize point allocation dynamically.
FR-005	Functional	The system shall publish final allocation results, generate individualized clash-free timetables for students, and export master	Medium	Pass: Downloadable PDF/ICS schedule produced with 0 slot clashes; Registrar roster matches enrolled	Closes registration lifecycle with actionable student schedules and

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		enrollment rosters for the Academic Registrar.		student IDs. Fail: Incomplete roster export or unassigned enrolled courses.	official registrar records.
NFR-001	Non-Functional (Performance)	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking confirms execution < 30 seconds for 5,000 concurrent bids. Fail: Execution exceeds 30 seconds or server runs out of memory.	Prevents server bottlenecks and ensures rapid turnaround.
NFR-002	Non-Functional (Security)	The system shall enforce Role-Based Access Control (RBAC) and maintain an immutable audit trail of all bid submissions, modifications, and administrative overrides.	High	Pass: Unauthorized role access blocked (403 Forbidden); 100% of bid changes logged with timestamp & user ID. Fail: Unauthenticated bid injection or unlogged state change.	Guarantees academic integrity and auditability during student grievance redressals.